

LAM NGUYEN

San Jose, CA • Inguyen0606@outlook.com • github.com/LamSauced • [linkedin.com/in/Inguyen0606](https://www.linkedin.com/in/Inguyen0606) • lamsauced.pages.dev

EDUCATION

California Polytechnic State University, San Luis

Expected May 2028

Obispo | *B.S. Computer Science*

- **Relevant Coursework:** Databases, Systems Programming, Data Structures & Algorithms, Computer Architecture.

De Anza College | *Computer Science & Engineering*

2024 – 2026

Coursework

- **Coursework:** Data Structures, x86 Assembly, MIPS Assembly, Linear Algebra, Differential Equations.
- **Activities:** Active member in ACM Club and 3D Printing / CAD Club; participated in collaborative hardware-software prototyping workshops.

TECHNICAL EXPERIENCE

Roku | *Intern Mentor*

San Jose, CA

Jan 2024 – Jul 2024

- Mentored ~30 incoming software engineering interns throughout the 2024 internship program, facilitating technical onboarding, architectural code reviews, and project guidance.
- Architected and deployed a full-stack React companion application integrated with interactive BrightScript development guides and project documentation to streamline intern onboarding.

Roku | *Software Programmer Intern*

San Jose, CA

Jun 2023 – Aug 2023

- Independently mastered BrightScript and SceneGraph frameworks to build ~80% of a production-ready Roku TV application for a company-wide technical showcase.

PROJECTS

Desktop AI Vision Assistant (LA Hacks) | *Python,*

LA Hacks

OpenCV, PySide/Qt, Multimodal LLMs, Local Fine-Tuning

- Engineered an interactive, floating-orb desktop overlay application in PySide/Qt that continuously captures screen context to deliver real-time AI suggestions with sub-second latency.
- Integrated screen capture pipelines with local and cloud multimodal LLMs; performed fine-tuning on LLaMA 7B models and deployed open-weight models (Gemma 2 27B, Qwen 2.5) using llama.cpp and Ollama.

Assistive Transcription Glasses & PoRTS | *Python, Linux*

Personal Project

(Arch/Ubuntu), CAVA, Bash, 3D CAD/Printing

- Architected a wearable assistive-technology prototype designed for Deaf and hard-of-hearing users, incorporating custom 3D-printed CAD enclosures and integrated circuitry.
- Developed a Linux/Python desktop companion app using Bash scripting for system service automation, featuring CAVA real-time audio visualization and automated timestamped note generation.

Piezoelectric Energy Harvesting Floor Tile | *C/C++,*

S4CA Showcase Award

Embedded Systems, CAD / 3D Printing

- Designed and prototyped a modular piezoelectric floor tile to convert kinetic energy into electrical power, owning embedded hardware integration and CAD design.
- Earned two monetary S4CA Product Showcase Awards for green energy innovation and proof-of-concept execution.

Low-Level Systems & Assembly Programming | *x86*

Academic Project

Assembly, MIPS, C, Linux CLI, GDB

- Designed assembly and C modules implementing low-level bitwise operations, custom bitmasking procedures, register manipulation, and memory-mapped I/O hardware device drivers.

PathPlanner, OpenCV, Limelight, Linux

- Programmed swerve drive control systems supporting field- and robot-relative movement modes using WPILib's command-based architecture.
- Configured Limelight/OpenCV target tracking and prototyped PhotonVision on an Orange Pi 5 single-board computer for autonomous trajectory planning and vision-guided scoring.

LEADERSHIP & TECHNICAL SKILLS

- **FRC Team 2643:** Programming Team Lead — Directed robot software architecture, structured codebases, and coached 10+ junior developers in Java, WPILib, and Git workflows.
- **AFA CyberPatriot:** Vice President & Windows Server Specialist — Hardened Windows Server systems, configured Active Directory security policies, and earned a Platinum State Round finish (highest score in state).

Languages & Frameworks: Java, Python, C/C++, JavaScript, TypeScript, SQL, Assembly (x86, MIPS), R, BrightScript, React, Astro, Node.js, HTML/CSS

Developer Tools & AI: Git, Linux (Arch/CachyOS, Ubuntu, Fedora), Bash Scripting, llama.cpp, Ollama, Fine-Tuning (LoRA), OpenCV, pandas, NumPy, Tidyverse, CAD/3D Printing, Windows Server Administration